

Digital Doppelganger – Basic Plan

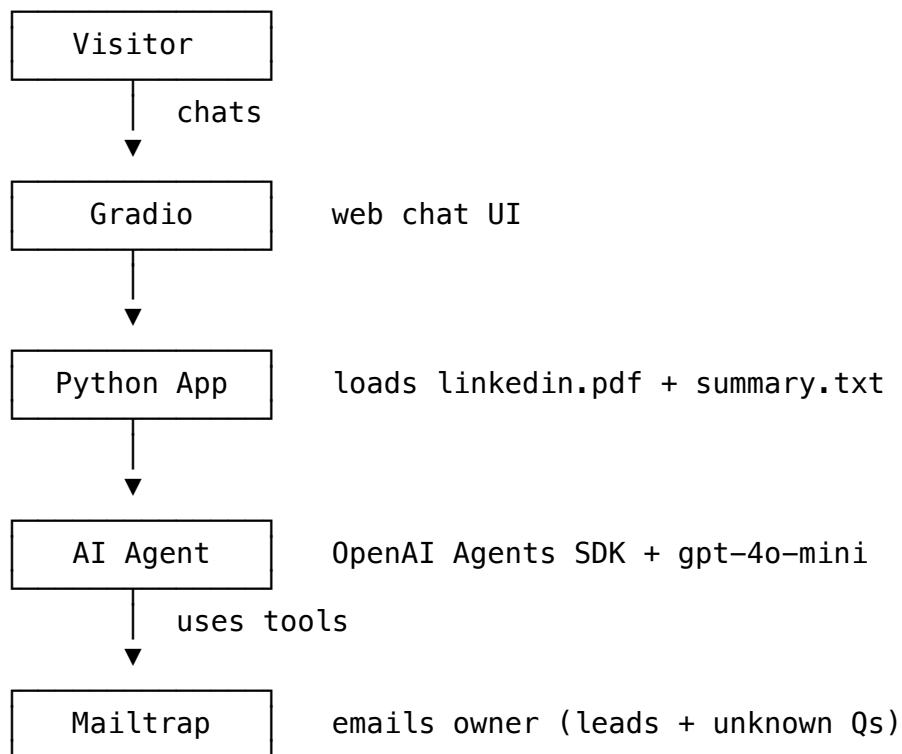
Project Summary

A web chatbot that role-plays as a specific person and answers questions about their career, background, and skills. It uses their LinkedIn profile (PDF) and a short bio as its source of truth. When a visitor shares their email, or asks something the bot can't answer, the bot emails the owner.

Built on the **OpenAI Agents SDK** (Python), structured around the four pillars of every harness: **Chat / Loop**, **Tools**, **Guardrails**, and **Observability**.

Tech Stack Flow Chart

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... Deployed on: Hugging Face Spaces

The Four Pillars

Whatever framework we use, these four responsibilities are always

present. Here is how
each maps onto this project and onto the OpenAI Agents SDK.

1. Chat / Loop – *drives reasoning across turns until the task is done*

- The OpenAI Agents SDK `Runner` drives the agent loop (model call → tool call → model call) until a final answer is produced.
- A single `Agent` is configured with instructions that establish the persona and embed the full source-of-truth documents (`linkedin.pdf` text + `summary.txt`) directly in the system prompt – no vector store / RAG given the small document size.
- Gradio holds the multi-turn conversation state and passes history back into the loop.

2. Tools – *lets the model read data and change the outside world*

- `record\_user\_details(email, name, notes)` – capture a lead and email the owner.
- `record\_unknown\_question(question)` – flag a question the bot couldn't answer and email the owner.
- Tools are defined with the SDK's `@function\_tool` decorator.

3. Guardrails – *constrains inputs, outputs, and actions to safe bounds*

Kept lightweight with inline checks (no separate validation agent/model for now).

- **Input guardrail:** stay on-topic (career / background / skills); deflect or refuse off-topic or unsafe requests – handled primarily via system-prompt instructions.
- **Output guardrail:** never invent facts not supported by the source documents; stay in persona – enforced through system-prompt instructions.
- **Action guardrail:** inline validation in tool code – check email format before triggering the Mailtrap tool; de-duplicate / rate-limit outbound emails.

4. Observability – *records what happened so you can debug and improve*

- Use the OpenAI Agents SDK's built-in **tracing** to record each run (model calls, tool calls,

- guardrail trips).
- Log conversations, tool invocations, and errors for later review.
- Track basic metrics: number of leads captured, unknown questions, refusals.

Open Questions / Next Steps

- Confirm model choice (`gpt-4o-mini` vs. another) and budget.
- Define the exact email templates Mailtrap will send.
- Confirm deployment target (Hugging Face Spaces) secrets handling for API keys.